

HELMET UNIT (TRANSMITTER)

```
#include <WiFi.h>
```

```
#include <esp_now.h>
```

```
// — Pins —————
```

```
#define IR_PIN 14 // IR sensor OUT → GPIO14
```

```
#define MQ3_D0_PIN 27 // MQ3 Digital D0 → GPIO27
```

```
#define MQ3_A0_PIN 34 // MQ3 Analog A0 → GPIO34
```

```
// — Bike ESP32 MAC (RECEIVER) —————
```

```
uint8_t receiverMAC[] = {0xC0, 0xCD, 0xD6, 0x85, 0x4B, 0x4C};
```

```
// — Alcohol threshold for analog reading —————
```

```
#define ALCOHOL_ADC_THRESHOLD 300
```

```
// — Data packet sent to Bike —————
```

```
typedef struct {
```

```
    bool helmetWorn;
```

```
    bool alcoholDetected;
```

```
    int alcoholADC;
```

```
    bool dataValid;
```

```
} DataPacket;
```

```
DataPacket data;
```

```
void onSent(const uint8_t *mac_addr, esp_now_send_status_t status) {
```

```
    Serial.println(status == ESP_NOW_SEND_SUCCESS ? "SENT OK" : "SENT FAIL");
```

```
}
```

```
void setup() {
```

```
Serial.begin(115200);
```

```
delay(1000);
```

```
pinMode(IR_PIN, INPUT);
```

```
pinMode(MQ3_D0_PIN, INPUT);
```

```
WiFi.mode(WIFI_STA);
```

```
delay(100);
```

```
Serial.print("Helmet MAC: ");
```

```
Serial.println(WiFi.macAddress());
```

```
if (esp_now_init() != ESP_OK) {
```

```
    Serial.println("ESP-NOW Init FAILED");
```

```
    return;
```

```
}
```

```
esp_now_register_send_cb(onSent);
```

```
esp_now_peer_info_t peerInfo = {};
```

```
memcpy(peerInfo.peer_addr, receiverMAC, 6);
```

```
peerInfo.channel = 0;
```

```
peerInfo.encrypt = false;
```

```
if (esp_now_add_peer(&peerInfo) != ESP_OK) {
```

```
    Serial.println("Add peer FAILED");
```

```
    return;
```

```
}
```

```
Serial.println("Helmet unit ready!");
```

```
}
```

```

void loop(){

  // — Read IR sensor —————

  // YOUR sensor: HIGH = helmet worn, LOW = not worn

  //  FIXED: was (== LOW), now (== HIGH)

  bool helmet = (digitalRead(IR_PIN) == LOW );


  // — Read MQ3 Digital (D0) —————

  // D0: HIGH = alcohol detected, LOW = clear

  bool alcoholDigital = (digitalRead(MQ3_D0_PIN) == HIGH);


  // — Read MQ3 Analog (A0) —————

  int alcoholADC = analogRead(MQ3_A0_PIN);

  bool alcoholAnalog = (alcoholADC >= ALCOHOL_ADC_THRESHOLD);


  // — Combine: either digital OR analog triggers —

  bool alcoholDetected = alcoholDigital || alcoholAnalog;


  // — Fill packet —————

  data.helmetWorn = helmet;

  data.alcoholDetected = alcoholDetected;

  data.alcoholADC = alcoholADC;

  data.dataValid = true;


  // — Send to Bike —————

  esp_now_send(receiverMAC, (uint8_t*)&data, sizeof(data));


  // — Serial debug —————

  Serial.print("IR RAW: ");

  Serial.print(digitalRead(IR_PIN));

  Serial.print(" | Helmet: ");

```

```
Serial.print(helmet ? "WORN" : "NOT WORN");  
Serial.print(" | Alcohol D0: ");  
Serial.print(alcoholDigital ? "YES" : "NO");  
Serial.print(" | ADC: ");  
Serial.print(alcoholADC);  
Serial.print(" | Alert: ");  
Serial.println(alcoholDetected ? "DETECTED" : "CLEAR");  
  
delay(500);  
}
```